Progress Report 3

Week 3: Baselines B1 & B2, Evaluation Engine, Beam Search, and Fusion Foundations

Cole Rodney Migisha	b29657	Charis Opol	b29654
Ainomugisha Faith	b27518	Lisa Lusinga Namugenyi	b30300
Ansasire Nehum Naamaye	b29785	Wakisa Isabelle	b30311

Project: LipScribe – Lip-Assisted Speech Recognition for Robust Live Captioning 5 November 2026

1 Summary

In Week 3 (30 October–5 November 2026), the team achieved all core objectives for Milestones **M3** and **M4**. Both unimodal baseline models (audio-only baseline **B1** and visual-only baseline **B2**) have completed full-corpus training on our dedicated Alienware GPU workstation and have been officially evaluated on the four held-out test speakers across all five standardised acoustic conditions (clean, 10, 5, 0, −5 dB).

The visual-only baseline (B2) demonstrates noise invariance across all SNRs, achieving a **23.04 % Test WER** and **14.03 % Test CER** on 3,998 test utterances, proving that the Conv3D–BiLSTM encoder generalises well to unseen speakers. In parallel, the dual-stream concatenation baseline (**B3**) has been fully implemented, verified via unit tests and synthetic overfit testing, and registered for training. On the decoding front, **CTC Prefix Beam Search** (W3-07) was implemented in numerically stable log-space, attaining a decode latency of 17.3 ms per utterance. Milestone M2 is fully closed, and Milestones M3 and M4 are satisfied on schedule.

4.16 %
B1 Clean WER (audio-only reference; 13.80 % at −5 dB)

23.04 %
B2 Test WER on unseen speakers (14.03 % CER; flat across SNR)

B3 Ready
Concat fusion model implemented and overfit verified

17.3 ms
CTC prefix beam search decode latency per utterance

2 Progress against the Milestones

Milestone	Due	Status and Progress
M1 Environment and dataset verified	Sun 25 Oct	**Done** – closed 29 Oct
M2 Preprocessing and alignment	Sun 1 Nov	**Done** – all 33 speakers cropped, normalised, parity verified
M3 & M4 Baselines B1, B2, B3	Sun 8 Nov	**Done** – B1 & B2 trained and tested; B3 implemented and tested
M5 Multimodal fusion (model P)	Sun 15 Nov	**In progress** – Cross-attention module design commenced
M6 & M7 Tuning and evaluation	Sun 22 Nov	Planned
M8 & M9 Application, testing, demo	Sun 29 Nov	Planned

3 Week 3 Task Deliverables

Table 1 details the status and verified outputs for each Week 3 work-breakdown task.

Table 1: Week 3 work breakdown cards and verified outcomes.

Card	Task	Owner	Status	Output and Evidence
W3-01	Complete B1 (audio)	Charis	Done	52 epochs; best epoch 43; WER 4.16 % (clean) to 13.80 % (−5 dB)
W3-02	Evaluation script	Lisa	Done	`scripts/evaluate.py`: per-speaker metrics, hypotheses CSV, gate hook
W3-03	Visual encoder V1	Cole	Done	3D-CNN + 2-layer BiLSTM; 5,246,368 trainable parameters; shape tests pass
W3-04	Visual augmentation	Faith	Done	Spatial crop jitter (±4 px) and horizontal flip; clip-synchronous
W3-05	Train B2 (visual-only)	Cole	Done	Overfit test (< 0.05 loss, 0.0 CER); trained 28 epochs; Test WER 23.04 %
W3-06	Implement B3 (concat)	Faith	Done	`fusion_concat.py`, `b3_concat.yaml`, `check_b3_overfit.py` pass
W3-07	CTC beam search	Isabelle	Done	Algorithm 2 in log-space; repetition handling; 17.29 ms decode latency
W3-08	Exp X5: audio features	Lisa	Planned	Deferred to Week 4 to prioritise baseline completion and fusion
W3-09	Experiment tracking	Nehum	Done	`history.csv`, live `learning_curves.png`, `plot_training_curves.py`
W3-10	Gradio Mode A skeleton	Isabelle	In progress	Frame capture and layout wired; demo integration in progress

4 Technical Highlights and Empirical Results

4.1 Comparative Baseline Performance: B1 vs. B2

Table 2 presents the official held-out test evaluation across the five standardised noise conditions for B1 and B2, decoding 3,998 test utterances from speakers `s1`, `s2`, `s20`, and `s22`.

Table 2: Official Baseline Evaluation: B1 Audio-Only vs. B2 Visual-Only on Held-Out Test Speakers.

Model Configuration	Test Word Error Rate (WER %) across Noise Levels (SNR)				
	Clean	**10 dB**	**5 dB**	**0 dB**	**−5 dB**
B1 Audio-Only (Log-Mel + BiLSTM)	**4.16 %**	5.47 %	6.49 %	8.69 %	13.80 %
B2 Visual-Only (3D-CNN + BiLSTM)	23.04 %	23.04 %	23.04 %	23.04 %	**23.04 %**

Key Insights:

- **Noise Invariance of Vision:** While B1 error degrades by a factor of 3.3× (from 4.16 % to 13.80 %) as acoustic noise intensifies, B2 maintains an unwavering 23.04 % WER and 14.03 % CER across all conditions.
- **Generalisation vs. Validation Discrepancy:** During validation training, B2 plateaued at 35.74 % WER due to speaker appearance calibration differences. When scored on the held-out test partition with four diverse speakers, performance improved to 23.04 % WER, confirming that the spatial 3D convolutions extract generalised viseme representations rather than overfitting to specific facial identities.

4.2 B3 Concatenation Model Architecture

To establish the unweighted multimodal baseline (W3-06), `B3Concat` combines frame-level representations from both encoders:

$$\mathbf{h}_t^{\text{concat}} = [\mathbf{a}_t\,;\,\mathbf{v}_t] \in \mathbb{R}^{1024}, \qquad \mathbf{f}_t = \text{Dropout}\big(\text{ReLU}(\text{LayerNorm}(\mathbf{W}_c\mathbf{h}_t^{\text{concat}} + \mathbf{b}_c))\big) \in \mathbb{R}^{256}. \quad (1)$$

The fused 256-dimensional features are passed directly into the standard 1-layer BiLSTM and CTC output projection head without attention or modality dropout. This enables a controlled ablation against Model P in Week 4.

4.3 CTC Prefix Beam Search Decoder

In accordance with Algorithm 2 of the implementation specification, prefix beam search maintains separate blank (p_b) and non-blank (p_{nb}) probabilities per candidate prefix $\ell \in \mathcal{P}$:

$$p_b'(\ell) \oplus\!= \log y_0^t + \big(p_b(\ell) \oplus p_{nb}(\ell)\big), \quad (2)$$

$$p_{nb}'(\ell\|k) \oplus\!= \log y_k^t + \big(p_b(\ell) + (1 - \delta_{k,\text{last}(\ell)})p_{nb}(\ell)\big). \quad (3)$$

Benchmarking demonstrates that the decoder achieves an average decode latency of **17.29 ms per utterance** on CPU, ensuring it easily satisfies the real-time constraint (< 40 ms) for live streaming captioning.

5 Risks, Resource Management and Next Steps

- **Workstation GPU Allocation:** GPU utilization on Alienware (RTX 4080 SUPER) has been smooth. Training B2 required ≈ 55 minutes for 28 epochs. B3 training is scheduled next.
- **Week 4 Focus (Model P):**
  1. Launch full B3 concatenation baseline training and evaluate on the test suite.
  2. Implement Model P with cross-attention fusion ($\mathbf{A} \to \mathbf{V}$ and $\mathbf{V} \to \mathbf{A}$) and learned per-frame gating g_t.
  3. Incorporate modality dropout ($p_a = 0.25, p_v = 0.15$) to force bidirectional reliance.
  4. Connect beam search with lexical dictionary constraints to push overall WER below 10 %.